

GARRISON, ND, USA

WMO: 727677

Lat: 47.646N

Lon: 101.439W

Elev: 1912

StdP: 13.71

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94041

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-14.8	-27.0	1.3	-19.6	-21.1	1.8	-14.4	36.6	16.3	32.9	17.2	7.7	300	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	90.1	68.5	86.2	67.9	82.6	66.1	73.0	83.2	70.7	81.7	68.7	79.3	11.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	117.6	79.1	67.1	106.9	76.6	65.0	99.2	73.9	38.0	82.6	35.8	81.5	34.0	79.5	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	26.1	22.8	DB	-27.2	97.1	5.8	3.9	-31.4	99.9	-34.7	102.2	-38.0	104.4	-42.2	107.3	(4)
(5)			WB	-26.6	77.3	4.2	3.3	-29.6	79.6	-32.1	81.5	-34.4	83.3	-37.5	85.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	11.6	12.4	26.7	41.0	53.5	63.1	68.8	67.5	58.6	44.1	28.8	16.0		
	DBStd	23.15	15.08	13.89	13.71	10.36	8.40	6.27	5.66	6.42	8.32	9.28	11.61	12.78		
	HDD50	5266	1189	1052	724	299	60	2	0	0	20	230	636	1054		
	HDD65	9053	1654	1472	1187	721	365	105	26	48	223	648	1085	1519		
	CDD50	2046	0	0	2	28	169	395	583	542	278	48	1	0		
	CDD65	358	0	0	0	0	9	48	144	125	31	1	0	0		
	CDH74	4158	0	0	0	22	172	539	1506	1410	475	34	0	0		
	CDH80	1373	0	0	0	4	37	127	508	542	149	6	0	0		
(14) Wind	WSAvg	10.1	10.6	10.0	10.7	10.9	11.1	9.8	8.1	8.4	9.3	10.6	10.7	10.3		
(15) Precipitation	PrecAvg	18.7	0.5	0.4	0.7	1.3	3.1	4.2	2.8	2.1	1.5	1.2	0.6	0.5		
	PrecMax	27.1	1.1	2.4	1.4	2.9	7.0	10.8	8.7	7.5	3.4	6.7	2.1	1.8		
	PrecMin	10.4	0.0	0.0	0.1	0.2	0.8	2.1	1.1	0.5	0.4	0.1	0.0	0.0		
	PrecStd	3.7	0.3	0.5	0.4	0.7	1.5	1.7	1.5	1.7	0.8	1.3	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	45.3	66.5	76.5	84.7	89.1	94.5	95.6	91.0	78.8	62.5	46.6		
		MCWB	38.1	38.3	49.5	54.9	61.4	68.8	70.3	69.0	65.6	58.7	49.2	39.1		
	2%	DB	39.2	39.2	57.4	69.6	78.7	83.3	89.5	90.5	84.1	70.9	54.3	40.1		
		MCWB	34.7	34.7	46.1	51.2	59.6	66.6	70.0	68.0	63.3	55.6	43.4	35.1		
	5%	DB	35.9	35.5	50.3	64.1	74.2	80.0	85.6	86.2	79.8	64.4	49.1	36.4		
		MCWB	32.8	32.2	42.4	48.5	57.3	64.8	70.2	66.9	61.6	51.3	41.3	32.8		
	10%	DB	32.2	32.1	43.4	58.9	69.9	76.6	82.0	81.9	74.8	59.0	44.1	33.3		
		MCWB	29.9	29.8	38.2	46.0	55.0	62.8	67.8	65.5	59.8	49.1	38.1	30.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	39.0	51.0	57.0	66.1	74.2	77.8	74.1	69.8	61.5	50.2	39.3		
		MCDB	44.5	44.4	63.7	73.6	77.0	82.9	87.2	86.3	83.0	75.2	61.2	45.7		
	2%	WB	35.0	35.1	46.4	52.4	62.5	70.2	74.3	71.2	66.8	56.5	45.0	35.4		
		MCDB	38.6	39.0	56.9	66.2	73.1	79.9	83.8	84.4	78.6	67.1	52.8	39.5		
	5%	WB	32.8	32.5	42.2	49.8	59.6	67.5	71.8	69.3	64.1	53.4	41.5	32.9		
		MCDB	35.8	35.1	49.9	62.1	69.8	76.0	81.5	82.1	75.0	63.1	47.9	36.1		
	10%	WB	30.0	29.9	38.5	47.0	57.0	64.7	69.6	67.6	61.3	50.1	38.6	30.4		
		MCDB	32.3	32.0	43.4	57.5	66.7	73.7	79.8	78.3	72.6	58.3	43.7	33.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	18.3	18.9	22.5	22.8	21.0	23.2	25.0	24.1	20.1	18.0	16.0		
		MCDBR	19.5	20.5	28.6	32.2	30.0	27.0	28.3	31.1	32.5	30.4	25.6	20.2		
		MCWBR	16.0	16.5	18.3	17.3	14.3	14.0	13.3	13.2	14.9	16.9	16.6	16.0		
	5% WB	MCDBR	18.8	19.7	27.0	30.6	24.0	22.8	24.1	26.3	27.6	28.0	24.5	19.6		
		MCWBR	15.7	16.2	18.0	17.3	12.9	14.0	13.3	12.4	13.9	16.4	16.4	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.259	0.286	0.324	0.344	0.366	0.374	0.381	0.336	0.302	0.267	0.247		
	taud		2.435	2.451	2.483	2.412	2.417	2.391	2.396	2.351	2.453	2.526	2.501	2.460		
	Ebn at Noon		271	294	300	295	291	284	280	273	277	273	263	259		
	Edh at Noon		20	25	29	34	35	36	36	36	30	23	19	17		
(44) All-Sky Solar Radiation	RadAvg		514	857	1223	1541	1822	1992	2106	1743	1302	814	499	366		
	RadStd		49	90	98	108	141	150	89	114	90	94	74	35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page